Introduction: Why Active Inference Needs an Auditable Learning Path

Problem: prerequisite opacity, numerical drift, and evidence-status confusion

Historical lineage and model boundary I

The Free Energy Principle is often situated in a historically bounded perception-as-inference context that includes Helmholtz's account of unconscious inference, Bayesian-brain accounts of neural uncertainty, and tutorials that make prediction error and variational bounds teachable (Helmholtz 1867; Knill and Pouget 2004; Bogacz 2017; Sprevak and Smith 2023). Its deeper background is selective rather than a single direct descent: ancient accounts of demonstration and ordered instruction, medieval optical work, early-modern method and probability, and eighteenth-century accounts of association and education all expose related questions about evidence, perception, uncertainty, and sequence (Aristotle 1901; Ibn al-Haytham 1989; Quintilian 95 AD; Bacon 1620; Descartes 1628; Huygens 1657; Newton 1704; Rousseau 1762). Bernoulli, de

Moivre, Bayes, and Laplace then anchor more specific histories of probability and inverse-probability reasoning, while Berkeley, Hume, Condillac, Hartley, Locke, Comenius, and Kant provide distinct traditions of perception, association, habit, experience, and structured cognition (Bernoulli 1713; Moivre 1718; Bayes and Price 1763; Laplace 1774; Berkeley 1709; Hume 1739; Condillac 1754; Hartley 1749; Locke 1690; Comenius 1657; Kant 1781). These sources are historical context and design analogies, not evidence that modern variational free energy, Bayesian brain models, or Active Inference already existed in their period; the fuller selective history appears in sec. 17.

Historical lineage and model boundary I

The modern FEP frames self-organizing systems in terms of variational free energy, with review work clarifying both the principle's scope and the need to avoid oversimplified readings (Friston et al. 2006; Friston 2010; Friston et al. 2023; Zhang and Xu 2024). In the discrete formulations used here, Active Inference supplies model-based rules for belief updating and policy evaluation: variational free energy is used for current inference, while expected free energy evaluates policies under stated preferences and observation models (Friston et al. 2017; Da Costa et al. 2020; Parr et al. 2022; Pezzulo et al. 2024). The word *agent* therefore names a modeled decision-making system; it does not by itself establish consciousness, intelligence, autonomy, or empirical adequacy.

Situation awareness: theory, model, software, and education are different objects

Education and delivery boundary I

The field is easiest to misread when four layers are collapsed into one label. At the theory layer, the FEP is a broad explanatory and mathematical framing whose scope, generality, and distinctive empirical predictions remain matters of active debate (Friston et al. 2023; Aguilera et al. 2022; Gershman 2019; Hodson et al. 2024). At the model layer, a discrete Active-Inference process theory specifies beliefs, likelihoods, preferences, policies, and an inference or planning scheme; changing those assumptions changes the model's behavior, even when authors use the same vocabulary (Friston et al. 2017; Da Costa et al. 2020; Parr et al. 2022; Millidge et al. 2021). A software implementation then adds a third layer: libraries such as `pymdp` and this project's kernels are executable realizations with different goals, tests, and coverage, not interchangeable evidence for the theory or for

biological adequacy (Heins et al. 2022; Oostrum et al. 2025).

The education layer adds a fourth boundary. Active-learning and prepared-environment proposals make the framework relevant to instruction, while knowledge-space and curriculum-aware cognitive-diagnosis work describe stronger learner-model commitments that require response data and an explicit account of learner state (Di Paolo et al. 2024; Doignon and Falmagne 1985; Falmagne and Doignon 2011; Fu and Fang 2025). Learning analytics can support such future evaluation, but its use also creates privacy, data-protection, validity, and human-oversight obligations (Liu and Khalil 2023; Sclater and Bailey 2015). SkillTree is a delivery substrate with its own runtime semantics; it does not convert a curriculum export into an intelligent tutor or turn a model demonstration into learner-outcome evidence (SkillTree 2026; National Security Agency 2020).

Education and delivery boundary I

A modeled Active-Inference agent is therefore a formal decision-making object with specified states, observations, preferences, and policies; it is not by itself an intelligent system. An intelligent system would require a separate capability and evaluation argument, while an adaptive learner-facing tutor would additionally require learner-state data, intervention logic, and prospective outcome evidence. Active Skillference occupies the intersection of these layers as a provenance-bound curriculum artifact: it makes a selected model family representable, teachable, and inspectable without claiming to settle the theory, personalize instruction, or measure learning.

Education and delivery boundary I

That compact summary hides a long prerequisite chain. To interpret expected free energy, a learner needs KL divergence; to interpret KL divergence, entropy and expectation; and to work with those quantities, probability, logarithms, optimization, and the geometry of distributions. A learner can encounter apex terms such as "expected free energy" or "epistemic value" before the mathematical foundations that make them derivable. The pedagogical risk is conceptual miscalibration: a learner may know the words before being able to compute the quantities or distinguish mathematical identities from process-theory commitments, empirical hypotheses, and philosophical scope claims. Active Inference tutorials and expected-free-energy derivation reviews show why the formal sequence cannot be replaced by slogans, while education-oriented and associative-learning Active Inference work frames learning as structured exploration of prepared

material, attention, and prediction-error reduction rather than passive term recognition (Smith et al. 2022; Millidge et al. 2021; Champion et al. 2026; Di Paolo et al. 2024; Anokhin et al. 2024).

A second problem is quantitative integrity. Hand-maintained numerical examples can drift from the equations they illustrate. Without an executable source of truth, a reader cannot tell whether a displayed value was computed, rounded, copied, or invented. This is a design risk that the platform addresses; it is not a prevalence claim about every tutorial or textbook.

A third problem is scholarship drift. A manuscript can accumulate citations that are syntactically valid but weakly connected to the claims they are supposed to support, while newer reviews or critiques change the evidential status of a phrase without changing its syntax. That matters here because technical critiques of FEP generality and recent empirical-status reviews, psychology-and-psychiatry progress reviews, and distinctive-prediction critiques belong in the evidential context (Aguilera et al. 2022, 2023; Hodson et al. 2024; Badcock and Davey 2024; Gershman 2019). This project treats the bibliography as part of the build contract: every cited source is mapped to a support lane and role, then checked before figures or learner artifacts are emitted.

Approach: provenance-bound prerequisites delivered through SkillTree

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Active Skillference addresses the three failure modes with one construction: a validated directed acyclic graph whose nodes are atomic *skills* and whose edges are explicit prerequisites. A skill is reachable only after the graph resolves the dependencies that precede it. The graph spans 8 strata from mathematical foundations to applications and is checked to be acyclic, connected from foundations to applications, and free of dangling prerequisites. This follows the basic premise of prerequisite-sensitive curriculum design: ordering is part of the learning object (Bloom 1956; Liang et al. 2018). The result is also interpretable through knowledge-space and learning-space theory, but only as an authored closure condition over course content; this release does not infer an individual's knowledge state or implement an adaptive diagnostic learning-space engine (Doignon and Falmagne 1985; Falmagne

and Doignon 2011).

The evaluation is correspondingly bounded by four questions. First, does the curriculum graph satisfy its declared structural and assessment invariants? Second, can quantitative examples be regenerated from tested kernels without unbacked or hand-entered result values? Third, can citation roles, manuscript claims, figures, and generated artifacts be audited as one evidence surface? Fourth, does the export preserve SkillTree's ownership of learning paths, quizzes, progress, and scoring? These questions evaluate the artifact and its delivery contract; they do not test whether the curriculum is optimal, whether an Active-Inference model is empirically true, or whether learners improve.

Approach: provenance-bound prerequisites delivered through SkillTree I

Every quantitative claim in a skill's content is produced by a tested computational kernel and inserted through a `claim:NAME` token. The platform refuses to publish a skill whose claims are not backed by code, and a linter flags any bare result number that slips into prose outside a claim token or LaTeX math. Thus the worked example that teaches entropy shows 0.693147 nats for a fair coin and 4.60517 nats for a one-in-a-hundred event because the kernel computed those values.

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Finally, the curriculum is built for delivery. It targets **SkillTree**, an open-source training and gamification platform whose public documentation describes Project -> Subject -> Skill composition, learning paths, and quiz-based knowledge checks (National Security Agency 2020, 2026b, 2026a; SkillTree 2026). Active Skillference maps subjects, skills, prerequisite learning paths, and quizzes onto that data model so the exported artifact can be inspected and, when the service is available, handed to a real progress-tracking environment. The citation is a platform-provenance claim: it identifies the SkillTree substrate and its government open-source origin, not an endorsement of this curriculum or a security assurance for any deployment.

Approach: provenance-bound prerequisites delivered through SkillTree I

The delivery layer also belongs to a wider community and open-science setting. The Active Inference Institute's public website repository describes a static resource hub with audience-specific pathways for newcomers, learners, researchers, developers, contributors, partners, and supporters; public repositories and learning resources; and an explicit boundary for sanitizing and verifying public projections (Active Inference Institute 2026). Active Skillference contributes one interoperable piece to that ecosystem: a provenance-bound curriculum and SkillTree delivery contract that makes formal examples, source roles, prerequisites, and release checks inspectable. This is a contribution to community-facing educational infrastructure, not a claim that the project represents the Institute, speaks for its contributors, or replaces its broader open-science programs.

Approach: provenance-bound prerequisites delivered through SkillTree I

These design choices turn the paper's contribution from a list of topics into a set of auditable contracts. Each contribution below is either a checked structure, a provenance gate, a source-role audit, or a delivery path into the existing SkillTree runtime.

The contribution map is therefore tied to six evidence classes. **Structural evidence** covers graph, prerequisite, assessment, and artifact invariants. **Computed evidence** covers values regenerated by tested kernels and resolved claim tokens. **Simulation evidence** covers fixed-input demonstrations whose behavior is conditional on an explicit model and parameter regime. **Literature evidence** covers historical lineage, formal definitions, pedagogical rationale, platform provenance, and critiques, each assigned a source role. **Delivery evidence** covers export, REST-contract, dashboard, and publication surfaces. **Scope-limit evidence** records what the validators and sources cannot establish:

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biological adequacy, learner-state inference, personalization, efficacy, retention, or transfer. No evidence class is used as a substitute for another.

Contributions: graph validation, provenance binding, scholarship audit, and delivery

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1. **Structural evidence:** a typed, validated **prerequisite-DAG engine** for curriculum design, including cycle detection, topological ordering, depth levels, and connectivity checks.
2. **Computed evidence:** a **content-provenance binding** mechanism that ties every learner-facing number to a tested kernel and fails the build otherwise.
3. **Curriculum representation:** an implemented **Active Inference curriculum** – 630 skills across all 8 strata – authored against that engine.
4. **Delivery evidence:** a **SkillTree exporter and seeder** that turns the curriculum into an importable, gamified training profile, with live availability treated as an external dependency.
5. **Literature evidence:** a **scholarship coverage audit** that maps manuscript citations to source roles and generates a reproducible coverage artifact.
6. **Scope-limit evidence:** an explicit boundary between formal model demonstrations, implementation checks, delivery checks, and future learner-outcome evaluation.